



Associations between built environment, perceived walkability/bikeability and metro transfer patterns

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ABSTRACT

Most studies that address the integration of active modes (i.e., walking and bicycling) and public transit tend to focus on either transfer distance or feeder mode choice. Few articles have reported the relationships between the built environment (D-variables) and transfer patterns (e.g., transfer distance and mode choice) in a single study. Furthermore, only very few studies accounted for travelers' judgements of walkability and/or bikeability and distinguished between access and egress trips. This paper fills these gaps in the literature using data collected in Nanjing, China. A random parameter Tobit and a random parameter multinomial logistic model are estimated to investigate respectively (i) the association of perceived walkability/bikeability and features of the built environment in metro catchment areas on walking/bicycling distance of access/egress trips and (ii) the probability of choosing active modes as transfer modes. The modeling results reveal that perceived walkability/bikeability and features of the built environment are more significantly correlated with mode choice behavior than with transfer distance. Moreover, walking/bicycling distance and access/egress mode choice tend to be strongly associated with perceived walkability and bikeability as well as with features of the built environment. The results are helpful in deepening urban planners and policymakers' understanding of how to design built environments that maximize the integration of active modes and metro.

1. Introduction

Many governments worldwide promote sustainable travel modes as a fundamental ambition of their transportation policies to alleviate the negative social and environmental externalities of increasing traffic congestion. Because metro systems are the natural backbone of sustainable transportation systems, stimulating travelers to shift from individual motorized transportation modes to rail transit and improving the position of rail transit are key challenges for transportation management and policymakers (Brons and Rietveld, 2009; de Souza et al., 2017; Li et al., 2020). Similar to other public transit systems, the use of a rail transit system involves an

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access trip from the origin to the boarding station and an egress trip from the alighting station to the destination. These access and egress stages are critical components of public transit trips and have a strong impact on the popularity of rail transit (Shelat et al., 2018). Access and egress trips can be made by different travel modes, either private or public, active or passive. Active modes (i.e.,

Table 1

Literature on associations between the built environment, perceived walkability/bikeability and transit transfer patterns.

Reference	Study region	Features of the built environment	Perceived walkability/bikeability	Dependent variable(s)	Transfer stage(s)
Böcker et al. (2020)	Oslo, Norway	Population density (+), Job density (+)	Not included	Propensity to use bicycle	Both access and egress
Chan and Farber (2020)	Toronto, Canada	Population density (/), Proportion of business/commercial area (–), Proportion of residential area (+), Proportion of industrial area (/), Intersection density (/), Street density (–)	Not included	Proportion of choosing active transport	Access
Guo and He (2020)	Shenzhen, China	Population density (/), Land use mix (+for morning egress and evening access), Proportion of residential area (+), Proportion of business/commercial area (/), Proportion of industrial area (+), Number of bus stops (+), Number of highways (/), Intersection density (–)	Not included	Bicycle usage	Both access and egress
Tao et al. (2020)	Twin cities, USA	Population density (–), Job density (+), Job and household entropy (+), Pedestrian network density (+), Intersection density (+)	Not included	Walking distance	Access
Wang et al. (2020)	Shenzhen, China	Population density (+), Intersection density (+), land use mix (+), Number of bus stops (/)	Not included	Bicycle usage	Access and egress merged
Lin et al. (2018)	Beijing, Taipei, Tokyo	Population density (+for Beijing), Job-housing balance (+for Taipei), Street length (– for Taipei), Arterial intersection density (+ for Beijing, – for Tokyo), Directness (– for Beijing and Taipei)	Not included	Bicycle usage	Access and egress merged
Ma et al. (2018)	Nanjing, China	Road density (–), Job density (–), Bus stop density (/), Resident density (/)	Not included	Bicycling activity space	Access and egress merged
Wu et al. (2018)	Nanjing, China	Not included	Sufficient walking/bicycling facilities (+), Comfort walking/bicycling environment (+), Enough walking/parking space (+)	Proportion of choosing walk or bicycle	Both access and egress
de Souza et al. (2017)	Rio de Janeiro, Brazil	Population density (–)	Cycle ways (+), Parking infrastructure (+), Current parking conditions (–), Public safety (–)	Propensity to use bicycle	Access
Ji et al. (2017)	Nanjing, China	Population density (+for walk), Number of bus lines (/), Number of parking slots (/)	Not included	Proportion of choosing walk or bicycle	Access
Wang and Cao (2017)	Twin cities, USA	Population density (/), Job density (–), Land use mix (+), Number of intersections (–)	Not included	Walking distance	Egress
Zhao and Li (2017)	Beijing, China	Number of malls (+ for walk, – for bicycle), Number of parks (+ for bicycle), Length of local roads (– for bicycle), Number of bus lines (– for both walk and bicycle)	Not included	Proportion of choosing walking or bicycling	Access and egress merged
Sun et al. (2016)	Beijing, China	Reach of road intersections (–), Floor area ratio (+), Reachable of buildings (–)	Difficulty to cross street (+), Pedestrian facilities (–), Continuity of sidewalks (/), Esthetic appeal of the pedestrian realm (–)	Walking time	Egress
Hochmair (2015)	Los Angeles, Atlanta, and Twin Cities	Intersection density (–)	Not included	Bicycling distance	Access
La Paix Puello and Geurs (2015)	South Wing of the Randstad area, Netherlands	Population density (/), Number of bus-metro lines (–), Average road quality (+)	Perceived connectivity (+), perceived quality of bicycle facilities (+)	Proportion of choosing bicycle	Access
El-Genaidy et al. (2014)	Montreal, Canada	Population density (–), Number of intersections (+)	Not included	Walking distance	Access
Jiang et al. (2012)	Jinan, China	Number of bus routes (/), Road length (/)	Protection, comfort, and enjoyment merged (+)	Walking distance	Access

Note: + indicates positive relationship; – indicates negative relationship; / indicates insignificant.

walking and bicycling) are considered green, economic, equitable and convenient due to their low cost, flexibility, positive physical and psychological health effects, and zero emissions (Gan et al., 2019). They are also recommended as desirable access and egress modes for public transit (Rietveld, 2000; de Souza et al., 2017). Consequently, the integration of metro and active access/egress modes is regarded as an effective way of reducing the negative externalities of car dependence and shaping sustainable urban structures (i.e., transit-oriented development).

“Sustainable urbanism” is a feasible way to promote the integration of active access/egress modes and metro and encourage green and healthy travel (Ding et al., 2018). It requires knowledge about the relationship between the built environment (D-variables, such as density, diversity, and design) and different aspects of transfer patterns. Consequently, the number of studies on the interdependencies of the built environment and travel behavior around metro stations has witnessed rapid growth (e.g., Jiang et al., 2012; El-Geneidy et al., 2014; de Souza et al., 2017; Zhao and Li, 2017; Lin et al., 2018; Zacharias and Zhao, 2018; Yang et al., 2019; Guo and He, 2020). Although, as a consequence, empirical evidence on the strength and nature of these interdependencies has been substantiated, nevertheless several gaps and imbalances in this literature can still be identified.

First, most studies focused on a specific aspect of transfer patterns, implying a more comprehensive understanding could not be developed. Thus, exploring the associations between features of the built environment and transfer patterns, considering both transfer distance and mode choice probability for access and egress trips, would enhance the value of these studies as this more comprehensive perspective could shed a light on the integration of active modes and metro from two perspectives: expanding catchment areas and encouraging travelers to switch away from other modes.

Second, the majority of existing studies were not only single-faceted in nature but also imbalanced in that they explored the associations of features of the built environment and walking transfer distance, but largely neglected bicycling transfer distance. This is unfortunate if it is realized that the bicycle is becoming a popular transfer mode to/from metro stations, reflected in the recent explosive growth of bike-sharing (e.g., Ji et al., 2017; Ma et al., 2018). Furthermore, the bicycle can significantly extend the catchment radius of a metro station, which implies that more people can potentially switch from other travel modes to the metro (Zhao and Li, 2017; Lin et al., 2019). Therefore, studying on the association between the built environment and both walking and bicycling distance would be helpful in better understanding possible differences between these two active modes in the context of access/egress trips to metro stations when changing particular features of the built environment.

Third, most prior studies only partly examined the association between the built environment and transfer patterns by considering either access or egress trips. Less attention has been paid to both the first and last mile in the same study (e.g., Yang et al., 2019; van Soest et al., 2020). Hence, there is a lack of understanding how associations between features of the built environment and transfer patterns vary between access and egress trips.

Fourth, compared to empirical evidence on associations between objective features of the built environment such as land use density and diversity, street connectivity, and distance to public transit stations, and travel patterns (e.g., Wang and Cao, 2017; Zhao and Li, 2017; Lin et al., 2018; Lin et al., 2019; Liu et al., 2020b), much less is known about the relationship between perceived walkability respectively bikeability and travel patterns in the context of metro transfer trips. The walkability and bikeability concepts refer to the extent walking respectively bicycling conditions are user-friendly (Nielsen and Skov-Petersen, 2018). Walkability and bikeability are difficult to define and measure objectively since different people may view the same conditions differently. Therefore, the concepts are usually using subjective components such as accessibility, safety, comfort, and attractiveness (Jiang et al., 2012; Sun et al., 2016; de Souza et al., 2017; Arellana et al., 2020). Detecting the relationships between aspects of perceived walkability/bikeability and transfer patterns can help policymakers identifying problematic walking/bicycling conditions and implement effective improvements to promote their integrated use. Moreover, neglecting perceived walkability/bikeability may exaggerate the influence of other factors, which in turn may lead to biased policy recommendations. In light of these considerations, this study aims to contribute to a comprehensive investigation of the associations between features of the built environment, aspects of perceived walkability/bikeability and transfer patterns to/from metro stations, considering both transfer distance and mode choice probability, using data collected in Nanjing, China. Therefore, this study addresses the gaps and imbalances in the current body of knowledge, signaled above: (1) it considers both transfer distance and choice of transfer mode; (2) it considers both walking and bicycling; (3) it examines the different associations in the context of both access and egress trips; (4) it includes perceived walkability and bikeability in the analyses.

The remainder of the paper is organized as follows. In Section 2, we review the literature related to our study. Section 3 presents the study area and describes the data, while the employed research methods are described in Section 4. Section 5 presents the estimation results. Section 6 provides the general discussion. The conclusions and recommendations for future studies are included in Section 7.

2. Literature review

This section provides a review of empirical findings and knowledge needs with regard to relationships between the built environment, perceived walkability/bikeability and transfer patterns, including transfer distance and mode choice behavior. In the following subsection, the literature on the association of the built environment and transfer patterns will be discussed. Next, the associations between perceived walkability/bikeability and transfer patterns will be reviewed. Table 1 provides a summary of selected, representative studies.

2.1. Associations between the built environment and transfer patterns

Numerous empirical studies have demonstrated the relationship between the built environment and transit transfer patterns.

Investigated features of the built environment include population density, land use diversity, street design, and accessibility to bus, and so forth.

2.1.1. Density

Sustainable urban design is founded on the contention that high density is instrumental in stimulating a shift toward active transportation modes. Logically, this contention is plausible in that higher density may imply short average distances, which in turn may stimulate the choice of active modes. An examination of the relevant literature about transfer distance, however, shows that the role of population and job density is rather inconclusive. For example, [El-Geneidy et al. \(2014\)](#) and [Tao et al. \(2020\)](#) found that walking distance to transit is negatively and significantly associated with population density around transit stations. By contrast, [Wang and Cao \(2017\)](#), and [Ma et al. \(2018\)](#) found that the relationship between population density and walking/bicycling egress distance is insignificant. Similarly, there is no consensus whether higher job density is associated with a shorter increases or bicycling distance ([Wang and Cao, 2017](#); [Ma et al., 2018](#); [Tao et al., 2020](#)).

Findings on the association between population/job density and transfer mode choice are also highly mixed. Empirical studies from Norway and China showed that population and job density are positively associated with active transfer mode use (e.g., [Ji et al., 2017](#); [Böcker et al., 2020](#); [Wang et al., 2020](#)). [de Souza et al. \(2017\)](#), however, found evidence of a significant negative association with bicycle choice for access trips in Rio de Janeiro, Brazil. Yet other studies reported non-significant associations for transfer trips ([La Paix Puello and Geurs, 2015](#); [Zhao and Li, 2017](#); [Guo and He, 2020](#); [Chan and Farber, 2020](#)).

2.1.2. Land use diversity

Land use diversity (mix) is a second cornerstone of New Urbanism. The contention is that mixed land use may lead to shorter distances between different types of land use. In turn, the shorter distances may allow traversing these distances on foot or by bike. Meanwhile, shorter distances may also lead to multi-stop, multi-purpose behavior. The potential mixed effects can also be found in the relevant empirical studies. Based on the data collected in Shenzhen, China, [Wang et al. \(2020\)](#) stated that high land use diversity in transit catchment areas led to shorter trips between departure/destination points and transit stations. On the other hand, two recent studies conducted in the Twin Cities, USA found that higher land use mix was associated with longer access and egress walking distance ([Wang and Cao, 2017](#); [Tao et al., 2020](#)).

Mixed land use may also encourage the use of active modes ([Sun et al., 2017](#)). [Zhao and Li \(2017\)](#), [Lin et al. \(2018\)](#) and [Guo and He \(2020\)](#) confirmed that areas with a more balanced land use mix around transit stations were associated with higher bicycle-transit integration. However, when focusing on specific land use (i.e., residential, commercial, and industrial land use), the conclusions become more mixed. While [Guo and He \(2020\)](#) found that the industry land use ratio was significantly and positively associated with bicycle use for egress trips during morning peak hours and for access trips during afternoon peak hours. [Chan and Farber \(2020\)](#) revealed a positive association between the residential land use ratio and active mode usage rates, while a negative association was found between the commercial/business ratio and active mode usage rates for trips integrating with transit in the Greater Toronto and Hamilton Area. In turn, [Zhao and Li \(2017\)](#) concluded that in Beijing the residential land use ratio was not significantly associated with integrated bike-metro use.

2.1.3. Street design

In the literature, relatively most interest has been focused on the number of intersections and road density. Again, their effects are not clear. On the one hand, a larger number of street intersections and higher road density indicate smaller block sizes and therefore better street connectivity and more route choice options ([Chan and Farber, 2020](#); [Liu et al., 2020b](#)). On the other hand, a larger number of intersections also mean a larger number of street crossings, reducing safety and pedestrian/bicycle flow. [Wang and Cao \(2017\)](#) found that the number of intersections was negatively related to walking egress distance, whilst [Hochmair \(2015\)](#) and [Guo and He \(2020\)](#) found that the number of intersections was negatively associated with bicycling access distance and bicycle-metro use. By contrast, [El-Geneidy et al. \(2014\)](#) and [Tao et al. \(2020\)](#) found evidence of a positive association of street intersection density and walking distance, while [Wang et al. \(2020\)](#) reported that street intersection density was positively associated with cycling frequency around the metro stations. In turn, a transnational comparative study, based on data from Beijing, Taipei, and Tokyo, showed that street intersection density was not significantly associated with bike usage for metro access ([Lin et al., 2018](#)). [Park et al. \(2014\)](#) arrived at the same conclusion, but did find that street intersection density was significantly associated with the probability of walking.

The evidence for road density is also mixed. Several studies found that road density was negatively related to bicycling distance to the metro station ([Zhao and Li, 2017](#); [Ma et al., 2018](#); [Chan and Farber, 2020](#)). However, [Guo and He \(2020\)](#) found it was not significantly related to bicycling around the metro stations.

2.1.4. Accessibility to bus

The choice of active modes does not only depend on the attractiveness of these modes, but also on the attractiveness of competing modes such as the bus. Previous studies have found that good accessibility to bus, measured by the number of bus stops or bus lines in metro catchment areas, negatively affects the choice to walk or bicycle to the metro. Thus, the substitution effect exists between buses and particularly bicycling. For example, [Zhao and Li \(2017\)](#) and [Liu et al. \(2020a\)](#) found that number of bus stops was negatively associated with bicycle- and walking-metro use. On the other hand, [Jiang et al. \(2012\)](#), however, found the relationship between number of bus lines and walk access distance was insignificant.

2.2. Associations between perceived walkability/bikeability and transfer patterns

Previous studies have noted that people who regularly walk and cycle are most concerned about external constraints such as presence of good walking facilities. In addition to objective features of the built environment, some scholars have argued that perceived walkability and bikeability play an important role in shaping intermodal behavior (e.g., Jiang et al., 2012; La Paix Puello and Geurs, 2015; Fernández-Heredia et al., 2016; Sun et al., 2016; Wu et al., 2018; Li et al., 2019; Arellana et al., 2020).

2.2.1. Perceived walkability

Residents living in the same physical environment may display a different willingness to walk to/from the metro station because they judge the walkability of the potential routes differently (e.g., Jiang et al., 2012; Li et al., 2019). Therefore, the association between perceived walkability and walking as a transfer mode has received growing attention in recent years. Sun et al. (2016), focusing on people living in high-density and compact-land neighborhoods, found that perceived characteristics such as pedestrian facilities, continuity of sidewalks, and difficulty to cross the street are significantly associated with people's walking time from metro stations, after controlling for socioeconomic characteristics, destination type, and features of the built environment. Similarly, Wu et al. (2018) found that sufficient perceived walking facilities and comfortable walking space were positively and significantly associated with the probability of choosing to walk for both access and egress trips. Arellana et al. (2020), reviewing urban walkability over the last ten years, concluded that pedestrians' perceived walkability (concerning pavement condition, sidewalk continuity, presence of pedestrian facilities) plays an important role in the decision to walk.

2.2.2. Perceived bikeability

Compared to the number of studies about perceived walkability, perceived bikeability has been investigated much less, particularly in the context of transit transfers. La Paix Puello and Geurs (2015) showed that perceived connectivity and quality of bicycle facilities are important in travelers' choice of bicycling as the access mode. Wu et al. (2018) found that perceived sufficient riding facilities, comfortable riding space, and safe parking spots are positively and significantly associated with the integration of metro and bicycle, after controlling for socio-demographics and trip attributes. Many studies indicated that shared bicycle users often fear the unavailability of shared bicycles when needed or the insufficiency of parking spaces when returning the bike, whilst private bicycle users usually care about the safety of the parking slots (La Paix Puello and Geurs, 2015; Fernández-Heredia et al., 2016; Ma et al., 2021).

2.3. Literature gaps and knowledge needs

Despite these valuable insights, four knowledge needs require particular attention, which motivated the authors to conduct this study. First, the association between features of the built environment and walking/bicycling transfer behavior shows significant variation across cities and countries. Moreover, only a limited number of studies included both the built environment and perceived walkability/bikeability when investigating transfer behavior. Second, although transfer distance and mode choice are two critical indicators, few authors explored distance and mode choice in a single study. Since expanding catchment areas and increasing probability of choosing active modes are two typical scenarios to promote the integration of active modes and transit, it is interesting to find out in which scenarios the built environment and perceived walkability/bikeability play more important roles. In other words, which scenario is more feasible and vigorous. Third, numerous studies only focused on one specific active mode. While a small number of studies included both walking and bicycling, these modes were regarded as a group. This may ignore behavioral differences between walking and bicycling. Fourth, most studies did not distinguish between access and egress stages. Therefore, there is a particular insufficiency in understanding how associations between the built environment, perceived walkability/bikeability and transfer patterns vary between the first and last mile (Shelat et al., 2018; Yang et al., 2019; Guo and He, 2020).

3. Research design and data

The data were collected in Nanjing, China, between October and November 2018. Nanjing is one of the cities in China where the metro is widely used after the completion of its first metro line in September 2005. By the end of 2018, ten metro lines were in operation with 161 stations and 378 km of tracks. The Nanjing metro system has been an essential transportation mode for serving the inner-city and connecting the city center and the periphery (Fig. 1). In 2018, annual ridership of the metro system exceeded 1.11 billion and the average daily ridership was about 3.05 million.

The survey was conducted both offline and online. The offline questionnaires were administered on a face-to-face basis and distributed during weekends in several urban parks and green spaces near metro stations. Mainly master students from the Department of Transportation Engineering, Southeast University were recruited to provide instructions to respondents during the face-to-face interviews. To ensure content validity of the survey, the wording, flow and explanations of the items were verified before the formal survey. Based on the feedback from volunteer interviewees, obvious problems were fixed and a final version of the survey version was prepared. In parallel with the face-to-face interviews, a popular Chinese online questionnaire survey platform named *Wenjuanxing* (<http://www.wjx.cn/>) was selected to conduct the online survey. To improve the reliability of the online survey, the IP (Internet Protocol) restriction technology was used. Only respondents with an IP address in Nanjing could get access the questionnaire.

The questionnaire consists of four sections. The first section includes questions about personal and household characteristics, such as age, gender, education level, vocation, personal monthly income, household size, and household vehicle ownership. The second section mainly includes questions about both access and egress trips, such as access/egress modes, transfer time, transfer distance, trip

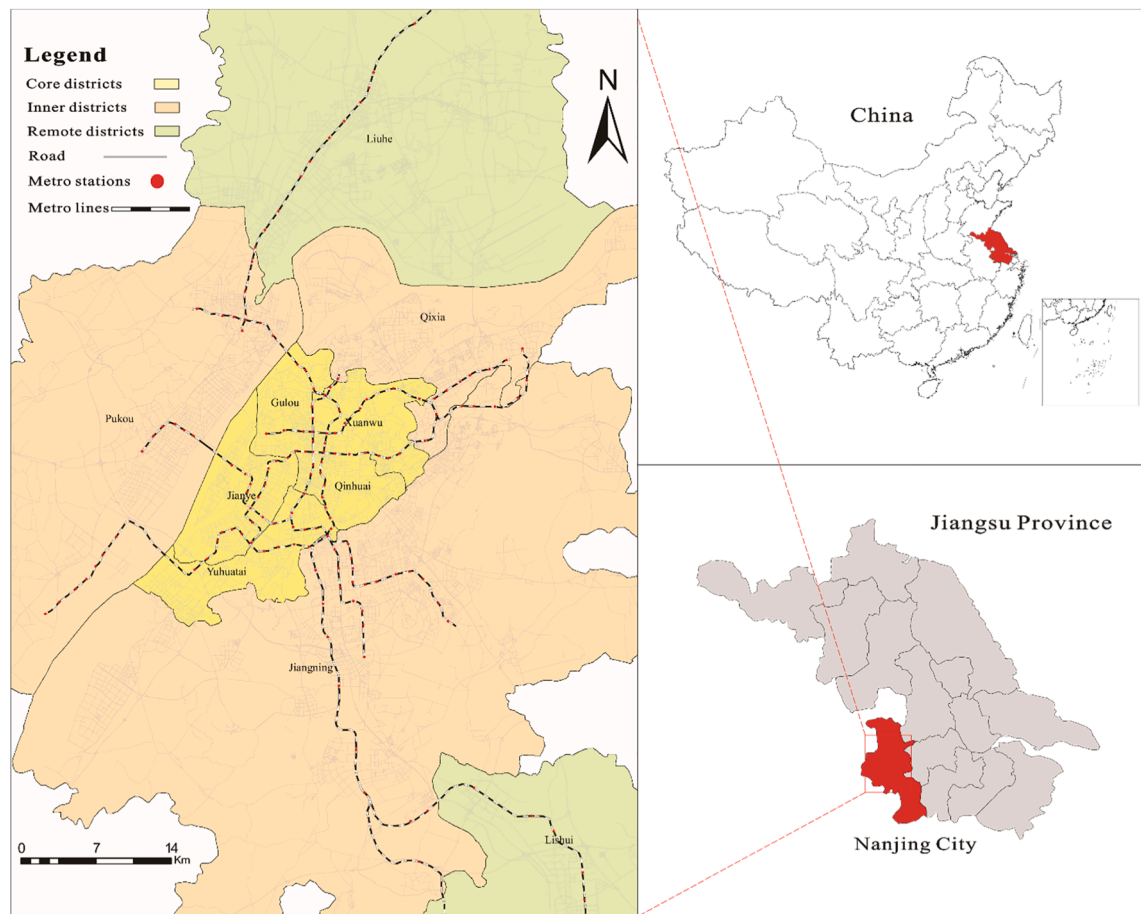


Fig. 1. Study area.

purpose (i.e., commuting, shopping, business), and stations that were entered and exited. The third and fourth sections concern the perceived walkability and bicycleability. Respondents were asked to judge walking and bicycling conditions of the access and egress trips on a 5-point Likert scale ranging from extremely disagree (1) to extremely agree (5). Following [Jiang et al. \(2012\)](#), [La Paix Puello and Geurs \(2015\)](#), [Sun et al. \(2016\)](#), [Wu et al. \(2018\)](#) and [Jamal et al. \(2020\)](#), the questions focused on accessibility, comfort, and attractiveness/enjoyment. The following questions were asked for each access and egress trip: (i) Do you agree that the current walking facilities such as pedestrian crossing, pedestrian traffic lights, sidewalk ground tile, and continuous sidewalk are good? (ii) Do you agree that the current walking space is sufficient (refer to the ease of walking including narrowed sidewalk due to hoardings, street cleanliness, street vending and disorderly parking in the sidewalk)? (iii) Do you agree that you achieve enjoyment because of the presence of activities and aesthetic aspects (i.e., the interaction between pedestrian and buildings, tree shade along with the sidewalk, and good eye-level greenness)? Similarly, three dimensions were used to evaluate bikeability: (i) Presence of good riding facilities such as presence of bicycle lanes, physical separation between vehicles and bicycles; (ii) Enjoyable riding environment related to aesthetics and relief from the elements (i.e., low emissions and noise from cars, shade from the sun, good eye-level greenness); (iii) Safe & enough parking space.

The built environment data were collected from various sources such as Baidu Map, OpenStreetMap, and the Nanjing Urban Planning Bureau (NUPB). Based on previous studies (e.g., [Sun et al., 2016](#); [de Souza et al., 2017](#); [Zhao and Li, 2017](#); [Aziz et al., 2018](#); [Lin et al., 2018](#); [Guo and He, 2020](#)), eight features of the built environment were selected: population density, proportion of three main land use types (residential, business/commercial, and industrial/manufacturing), land use mix, number of intersections, road density, and number of bus lines ([Table 2](#)). The land use CAD DWG file of the Nanjing region from the NUPB was used to determine the proportions of the three main land use types. Land use mix was calculated in terms of the commonly used entropy measure ([Cervero](#)

Table 2
Description of variables and descriptive statistics.

Variable	Description	Frequency	Percentage		
<i>Socio-demographics</i>					
Gender	Male	364	48.66%		
	Female	384	51.34%		
Age	<19	60	8.02%		
	19–25	336	44.92%		
	26–45	282	37.70%		
	>45	70	9.36%		
Education level	High school or lower	84	11.23%		
	Undergraduate	499	66.71%		
	Master or above	165	22.06%		
Job type	Student	314	41.98%		
	Government or Enterprise	330	44.12%		
	Self-employed	104	13.90%		
Working hours (per day)	≤8 h	436	58.29%		
	>8 h	312	41.71%		
Personal monthly income	<4000 <i>yuan</i>	350	46.79%		
	4000–8000 <i>yuan</i>	255	34.09%		
	> 8000 <i>yuan</i>	143	19.12%		
Household size	≤3 persons	519	69.38%		
	>3 persons	229	30.61%		
Preschooler (<6 years old)	No	585	78.21%		
	Yes	163	21.79%		
Household non-motorized vehicle ownership	No	241	32.22%		
	Yes	507	67.78%		
<i>Trip characteristics</i>		<u>Access stage</u>		<u>Egress stage</u>	
		Frequency	Percentage	Frequency	Percentage
Peak hours	No	432	57.75%	490	65.51%
	Yes	316	42.24%	258	34.49%
Travel mode	Walking	380	50.80%	608	81.28%
	Bicycling	182	24.34%	85	11.36%
Trip purpose	Passive travel modes	186	24.86%	55	7.35%
	Commuting or go to school	322	43.05%	322	43.05%
	Others	426	56.95%	426	56.95%
Urban (metro station location)	Yes	225	30.08%	437	58.42%
	No	523	69.92%	311	41.58%
Access/egress distance (km) ^a	1.70	2.02	0.87	1.20	
<i>Perceived walkability and bikeability</i>		<u>Access stage</u>		<u>Egress stage</u>	
		Mean	Std. dev.	Mean	Std. dev.
Walking	Presence of good walking facilities	4.23	0.87	4.28	0.72
	Ease of walking	4.08	0.91	4.14	0.83
	Enjoyable walking environment	3.97	0.95	4.09	0.82
Bicycling	Presence of good riding facilities	4.01	0.96	3.99	0.91
	Enjoyable riding environment	4.00	0.91	3.95	0.95
	Safe & enough parking space	3.72	1.04	3.82	0.98
<i>Built environment</i>					
Population density (persons/km ²)	12,897	11,228	25,019	16,992	
Residential ratio	0.32	0.15	0.29	0.13	
Business/commercial ratio	0.10	0.13	0.06	0.11	
Industrial/manufacturing ratio	0.11	0.13	0.27	0.23	
Land use mix (entropy)	0.50	0.12	0.61	0.18	
Number of intersections	14.16	7.67	21.18	9.53	
Road density (km/km ²)	10.16	2.67	12.12	3.13	
Number of bus lines	11.74	6.95	12.74	6.50	

^a The mean and standard deviation are presented here.

and Kockelman, 1997; Ding et al., 2017; Gan et al., 2020). The number of intersections and road density were acquired from OpenStreetMap, while the number of bus lines was collected from Baidu Map. All features of the built environment were calculated based on a 1.5 km radius around the metro station using geographic information system ArcGIS version 10.3. Five main travel modes were identified for each access and egress trip: walking, bicycle, bus, taxi, and car-hailing. In this study, we grouped bus, taxi, and car-hailing into the category ‘passive travel modes’¹.

In total, 827 questionnaires were distributed, of which 121 online. After excluding questionnaires with missing answers, 748 usable questionnaires were retained. An overview of the sample is presented in Table 2. The proportion of male and female is similar to the corresponding distribution of the 2019 Nanjing Statistical Yearbook (Nanjing Statistical Bureau, 2019). In terms of age, the majority of the respondents are between 19 and 45 years old, accounting for more than 80%, which is similar to previous studies in Nanjing (He et al., 2018; Wu et al., 2018). Most respondents are highly educated (about 67% graduated from a junior college or university, while 22% has a Master’s degree). In part, this high percentage may be explained by the fact that Nanjing is one of main educational centers and most developed areas in China. Many of the highly educated people have the opportunity to find a job in the area. More than 40% of the respondents are students, which means that students are overrepresented. Nanjing with a total population of 8.5 million has 2.07 million students, of which more than a half are college students (Nanjing Statistical Bureau, 2019). The share of personal monthly income decreases from >8,000 yuan to <4,000 yuan, accounting for respectively 47%, 34%, and 19%, while the 2018 mean per capita annual income for Nanjing is about 59,000 yuan. Moreover, 69% of the respondents belong to a household of less than 3 persons, while 31% belong to a household with more than 3 persons against a 2018 average household size of 2.79 (Nanjing Statistical Bureau, 2019).

Thus, as expected, the sample is not representative of the Nanjing population. More importantly, the sample should be representative of the population using the metro system. Unfortunately, we cannot judge it is because the official data of metro users are not publicly available. Nevertheless, since the purpose of this paper is not to forecast integrated use, but to examine associations between the built environment, perceived walkability/bikeability and transfer patterns, any lack of sample representativeness is not a major issue as it is unlikely it would substantially affect the results of the model estimations.

4. Methodology

The general aim of this study is to examine (i) the relationship between features of the built environment, respectively perceived walkability/bikeability and transfer distances, and (ii) the relationship between features of the built environment, respectively perceived walkability/bikeability and the probability of choosing active modes for both metro access and egress trips, after controlling for socioeconomic characteristics and trip attributes. Transfer distance is continuous, non-negative and regarded as a variable left-censored at zero, Tobit model is employed. Because access/egress mode choices are categorical variables, a multinomial logistic regression model is applied. Finally, to account for unobserved heterogeneity across individuals, random parameter versions of these models are estimated.

4.1. Random parameters Tobit model

The Tobit model assumes that the dependent variable Y_i is equal to the latent variable Y_i^* when Y_i^* is positive, while to zero when Y_i^* is less than or equal to zero. The random parameters Tobit model proposed for the analysis can be described as follows:

$$Y_i^* = \alpha_i X_i + \varepsilon_i \quad (1)$$

$$Y_i = \begin{cases} Y_i^* & \text{if } Y_i^* > 0 \\ 0 & \text{if } Y_i^* \leq 0 \end{cases} \quad (2)$$

$$\alpha_i = \alpha + \sigma_i \quad (3)$$

where Y_i^* denotes the latent variable linking the expected walking/bicycling distance for i th traveler, Y_i is the observed walking/bicycling distance for i th traveler, X_i is the vector of alternative specific independent variables for i th traveler (pedestrian or bicyclist), ε_i is the error term, α_i represents the vector of estimated parameters for i th traveler, α denotes the vector of the mean (or fixed) estimated parameters, and σ_i represents the vector of randomly distributed terms such as a normal distribution.

4.2. Random parameters multinomial logistic regression model

The random parameter multinomial logistic regression model was employed to identify factors affecting active mode choice for access and egress trips, accounting for unobserved heterogeneity. Assuming P represents the probability of using a specific mode as the transfer mode, a random parameter multinomial logistic regression model for choosing walking respectively bicycling as the transfer

¹ The number of observations of taxi, car-hailing and bus for access and egress trips are respectively small. It is unwise to make a much-detailed travel mode split for modelling transfer mode choice based on the restricted sample size. Meanwhile, the primary goal of this study is to address the associations between independent variables and active transfer behavior. So, we grouped taxi, car-hailing and bus into one single group of passive transport modes.

mode can be expressed as follows:

$$\ln\left(\frac{P(\text{walking})}{P(\text{passive modes})}\right) = \beta_{1i}x_{1i} \quad (4)$$

$$\ln\left(\frac{P(\text{bicycling})}{P(\text{passive modes})}\right) = \beta_{2i}x_{2i} \quad (5)$$

$$\beta_{ji} = \beta_j + \sigma_j\omega_{ji}, j = 1, 2 \quad (6)$$

where passive modes mean any mode (i.e., taxi, bus, and car-hailing) other than walking and bicycling, x_{ji} is the vector of alternative specific independent variables for j th alternative of i th traveler, β_{ji} denotes the vector of parameters estimated for j th alternative of i th traveler, β_j denotes the vector of the mean (or fixed) estimated parameters for j th alternative, σ_j is the vector of standard deviations for the random parameters of j th alternative, and ω_{ji} represents the vector of randomly distributed terms such as a normal distribution. When excluding the term $\sigma_j\omega_{ji}$ in Eq. (6), the random parameters logistic regression model degenerates into the classical (fixed parameter) logistic regression model.

Table 3

Random parameters Tobit regression analysis for access/egress distances of active modes.

Variables	Walking					Bicycling				
	Coef.	Std. Error	2.5%	97.5%	Elasticity	Coef.	Std. Error	2.5%	97.5%	Elasticity
<i>Built environment</i>										
Population density	-4.23×10^{-6}	2.38×10^{-5}	-4.25×10^{-5}	5.09×10^{-5}	-0.10	-6.58×10^{-5}	7.09×10^{-5}	-2.04×10^{-4}	7.32×10^{-5}	-0.05
Residential ratio	-0.165	0.088	-0.337	0.008	-0.74	-0.078	0.317	-0.700	0.543	-0.67
Business/commercial ratio	-0.085	0.101	-0.283	0.113	-0.28	0.423	0.440	-0.440	1.287	0.31
Industrial/manufacturing ratio	-0.259*	0.100	-0.455	-0.064	-2.93	0.053	0.376	-0.683	0.790	0.36
Std. dev. of parameter distribution	0.251*	0.063	0.127	0.375						
Land use mix	0.417*	0.092	0.236	0.598	1.37	0.058	0.364	-0.772	0.656	0.23
Std. dev. of parameter distribution	0.177*	0.014	0.149	0.205						
Number of intersections	-0.004	0.003	-0.009	0.002	-0.11	-0.011	0.012	-0.035	0.013	-0.10
Road density	0.001	0.006	-0.011	0.014	0.02	0.023	0.024	-0.023	0.069	0.15
Std. dev. of parameter distribution						0.065*	0.003	0.003	0.127	
Number of bus lines	-0.005*	0.002	-0.008	-0.002	-0.96	-0.004	0.009	-0.022	0.015	-0.26
<i>Perceived walkability/bikeability</i>										
Presence of good walking facilities	-0.048*	0.015	-0.078	-0.017	-3.10	n.a.				
Ease of walking	0.036	0.027	-0.037	0.069	1.24	n.a.				
Enjoyable walking environment	-0.032*	0.016	-0.063	-0.001	-1.96	n.a.				
Presence of good riding facilities	n.a.					0.094	0.054	-0.037	0.242	2.54
Enjoyable riding environment	n.a.					-0.192*	0.059	-0.307	-0.076	-3.41
Safe & enough parking space	n.a.					-0.010	0.046	-0.102	0.081	-0.25
<i>Trip attributes</i>										
Access trip	0.113*	0.022	0.070	0.156	30.63	0.550*	0.088	0.377	0.723	34.20
Std. dev. of parameter distribution	0.130*	0.014	0.102	0.158		0.951*	0.037	0.878	1.024	
Urban	-0.089*	0.028	-0.144	-0.034	-20.16	-0.034	0.102	-0.167	0.235	-3.29
Peak hours	0.022	0.022	-0.021	0.065	5.19	0.047	0.079	-0.107	0.201	4.76
Trip purpose	0.003	0.022	-0.040	0.048	0.35	0.105	0.080	-0.051	0.262	3.28
Intercept	0.829*	0.091	0.651	1.007		0.985*	0.377	0.247	1.724	
<i>Socioeconomic attributes</i>										
Yes						Yes				
$\ln L(0)$	-438.225					-339.147				
$\ln L(\hat{\beta})$	-356.046					-294.589				
Pseudo R ²	0.187					0.131				

Notes: ‘*’ denotes $p < 0.05$, significant at the 95% level; ‘n.a.’ manifests that this independent variable is not included in the model.

Table 4

Random parameters logistic regression analysis for transfer mode choice in access and egress trips.

Variables	Access trip						Egress trip					
	Walking			Bicycling			Walking			Bicycling		
	Coef.	Std.Error	Elasticity	Coef.	Std.Error	Elasticity	Coef.	Std.Error	Elasticity	Coef.	Std.Error	Elasticity
<i>Built environment</i>												
Population density	0.00037*	0.00016	0.22	0.00011	0.00023	0.04	0.00079*	0.00035	0.36	−0.00048	0.00038	−0.08
Residential ratio	−1.804	2.458	−0.07	−1.279	0.957	−0.02	−5.346*	2.793	−0.63	−1.607	1.752	−0.10
Business/commercial ratio	8.677*	3.919	0.68	−3.096	2.003	−0.21	3.032	2.012	0.25	2.754	2.313	0.28
<i>Std. dev. of parameter distribution</i>	5.036*	2.385										
Industrial/manufacturing ratio	−2.237	1.542	−0.13	−1.989	1.230	−0.08	−0.237	1.641	−0.02	−1.582	2.139	−0.09
Land use mix	1.343	0.862	0.20	4.682*	1.340	0.74	3.738*	1.388	0.62	4.432*	1.113	0.75
<i>Std. dev. of parameter distribution</i>				2.513*	1.653		2.133*	0.821		3.083*	1.382	
Number of intersections	0.082	0.040	0.15	0.078*	0.032	0.38	0.065	0.240	0.12	0.099*	0.044	0.41
Road density	−0.021	0.067	−0.04	−0.040	0.079	−0.05	−0.284	0.206	−0.10	−0.261	0.172	−0.08
Number of bus lines	−0.069	0.043	−0.16	−0.155*	0.032	−0.47	−0.053	0.037	−0.13	−0.139*	0.026	−0.46
<i>Perceived walkability/bikeability</i>												
Presence of good walking facilities	1.683*	0.774	1.65	n.a.			1.852*	0.650	1.75	n.a.		
Ease of walking	1.729*	0.849	1.84	n.a.			0.763	0.638	0.32	n.a.		
Enjoyable walking environment	1.389*	0.707	0.94	n.a.			1.166*	0.506	1.28	n.a.		
Presence of good riding facilities	n.a.			1.716*	0.758	1.36	n.a.			1.399*	0.502	1.55
Enjoyable riding environment	n.a.			1.096*	0.506	0.75	n.a.			1.105*	0.502	0.94
Safe & enough parking space	n.a.			1.273*	0.620	0.88	n.a.			1.178*	0.448	1.36
<i>Trip attributes</i>												
Urban	0.218	0.185	2.59	0.170	0.108	1.62	0.225	0.173	2.57	0.149	0.086	1.24
Peak hours	−0.376	0.217	−4.12	−0.223	0.252	−8.27	0.455	0.373	6.39	0.186	0.438	4.38
Trip purpose	0.385	0.229	9.68	0.593*	0.301	13.31	0.376	0.854	5.38	0.383*	0.153	12.72
<i>Std. dev. of parameter distribution</i>				0.632*	0.287							
Intercept	0.744	0.658		−1.257*	0.508		0.857	0.463		−1.363*	0.546	
Access/egress distance	Yes						Yes					
<i>Socioeconomic attributes</i>	Yes						Yes					
$\ln L(0)$	−755.845						−533.168					
$\ln L(\hat{\beta})$	−194.483						−189.808					
Pseudo R ²	0.742						0.644					

Notes: ‘*’ denotes $p < 0.05$, significant at the 95% level; ‘n.a.’ manifests that this independent variable is not included in the alternative.

Simulated maximum likelihood techniques are used to estimate the random parameter models using NLOGIT version 6.0. This software uses Halton sequences to estimate the models. Comparison of log-likelihoods values at zero and at convergence, and pseudo R^2 are used to evaluate the goodness-of-fit of the models.

5. Results

The random parameters can be specified in terms of various prior distributions (e.g., normal, uniform, and log-normal). The random parameters are assumed to obey the normal distribution due to the good performance finally. This assumption also allows the models to capture the mixed results reported in the literature review. The random parameter was kept in the model only when its estimated σ_i^2 was significantly larger than 0, otherwise a fixed value was used. The reported estimations are based on 500 Halton sequences. The parameter estimates and model fit information for the Tobit and logistic models are shown in Tables 3 and 4. The pseudo R^2 of the distance models are both lower than 0.20, indicating a modest model fit. The pseudo R^2 of the mode choice models is 0.742 and 0.644 respectively, indicating a high goodness-of-fit. In order to better evaluate the association between the built environment, perceived walkability/bikeability and transfer patterns, we also calculated the (pseudo) elasticity of each independent variable (Washington et al., 2010).

5.1. Features of the built environment

5.1.1. Density

It is found that population density has a significant positive association with the probability of choosing walking as the transfer mode in conjunction with metro for both access and egress trips. It is not significant for the probability of choosing bicycling as the transfer mode. This indicates that residents are inclined to select walking rather than the other modes (bicycling, bus, and taxi, etc.) for access/egress trips when the population density within the metro station catchment area is high, which is consistent with the findings of Ji et al., (2017). With a 1% increase in population density, the elasticity indicates that the probability of choosing walking increases by 0.22% and 0.36% for respectively access and egress trips.

As for transfer distance, consistent with earlier findings (Wang and Cao, 2017; Ma et al., 2018), population density is found not to be significantly associated with either walking or bicycling distance. Given that Nanjing is a densely populated city with 8.50 million citizens covering 1125 km² urban built-up area, it is consistent with findings in the Twin cities case (Tao et al., 2020), which corroborates that walking distance becomes stable after 18 people/acre.

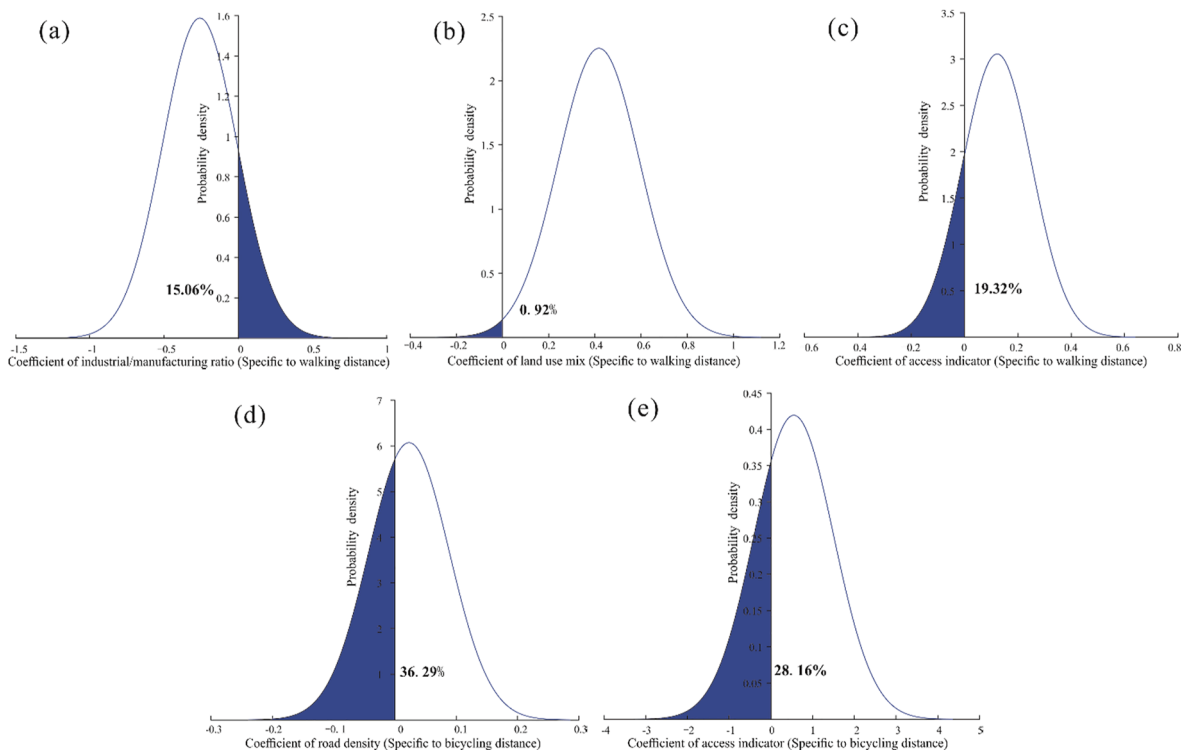


Fig. 2. Estimated parameter distributions of variables in transfer distance modelling.

5.1.2. Land use diversity

The results displayed in Table 3 show that the industrial/manufacturing ratio has a significantly negative effect on walking distance indicating that the higher the industrial/manufacturing ratio, the shorter the distance that people walk. With a 1% increase in the industrial/manufacturing ratio, the elasticity indicates that the walking distance decreases by an average of 2.93%. This result is in accordance with the reality that industrial zones are normally large and enclosed, resulting in poor street connectivity and accessibility for walking. Furthermore, the noise and heavy air pollution from industrial zones, and poor walking conditions (many trucks and lack of sidewalks) may reduce the possibility of walking to or from the nearby metro stations. The parameter of industrial/manufacturing ratio in the walking distance model is normally distributed with mean -0.259 and standard deviation 0.251 (Table 3). The corresponding plot (Fig. 2(a)) indicates that an increase in land use mix will decrease the walking distance of most pedestrians (84.94%) but increase walking distance for 15.06% of the pedestrians. The relationships between the other two selected land use variables (residential ratio and business/commercial ratio) and walking/bicycling distance are insignificant.

Table 4 shows that the associations between land use variables and active mode shares are a little more complex. Residents are less likely to choose walking in the context of egress trips when the metro stations are located in an area with a high residential ratio, which is inconsistent with the findings in the Greater Toronto and Hamilton Area, Canada (Chan and Farber, 2020). This is somewhat counterintuitive but may be largely explained by the current community layout in Chinese cities. The communities and neighborhoods are usually centralized, large and enclosed resulting in low block connectivity and bad walking accessibility. Thus, people have to take a detour to walk from a metro station to their home. However, the association between residential ratio and the probability of choosing bicycling is not significant. The business/commercial ratio is found to be positively associated with the probability of choosing walking in the context of access trips, indicating that travelers are prone to walk to the metro stations when this ratio is high. The significant standard deviation of this parameter (5.036, Table 4) indicates that the effect of the business/commercial ratio on mode choice in the context of access trips varies across individuals. A higher business/commercial ratio would stimulate most travelers (95.76%, see Fig. 3(a)) to choose walking for access trips. The models suggest that the business/commercial ratio is, however, not significant for the probability of choosing bicycling as the transfer mode for both access and egress trips. As for the industrial/manufacturing ratio, the models estimate a negative effect on the probability of choosing an active transfer mode (both walking and bicycling), regardless of the transfer stage. But the effects are quite modest, as indicated by the signs of the corresponding coefficients.

In contrast to the complicated relationships between transfer patterns (distance and choice probability) and specific land use variables, land use mix is found to be positively related to both distance and choice probability, regardless the active mode is walking or bicycling. This is consistent with the findings of Wang and Cao (2017) and Guo and He (2020), who found that mixed land use is significantly related to both walking distance and bicycle usage. The elasticity indicates that a 1% increase in the land use mix will increase walking distance by 1.37%, and the probability of bicycling by 0.74% and 0.75%, for access and egress trips respectively. On the other hand, the parameter of land use mix in the walking distance model is normally distributed with mean 0.417 and standard deviation 0.177 (Table 3). The distribution of land use mix is illustrated in Fig. 2(b). It indicates that an increase in land use mix will increase the walking distance of most pedestrians (99.08%) but decrease walking distance for 0.92% of the pedestrians. This result confirms the unobserved heterogeneity effect where individuals have different reactions to the built environment. Besides, based on the corresponding sign and elasticity, the positive association between bicycling distance and land use mix is modest.

The significant positive coefficient of land use mix in Table 4 indicates that travelers show an increased propensity to use active travel modes (i.e., walking, bike) rather than passive travel modes (i.e., bus, car) for access/egress trips. Furthermore, the elasticity analysis shows that a 1% increase in land use mix results in a 0.62% and 0.75% increase in the probability of respectively walking and bicycling in the context of egress trips from metro stations. This is consistent with the findings of a meta-analysis by Ewing and Cervero (2010) and some recent empirical studies (i.e., Ding et al., 2017, 2018; Tao et al., 2020). Moreover, the models estimate normally distributed random parameters for land use mix effect. The plotted normal curve shows that, in the context of egress trips, the association is negative for 3.98% of the population – higher land use mix reduces the likelihood of choosing walking. This maybe because of possible heavy traffic, a complex neighborhood environment and more mode alternatives (e.g., sharing bike, taxi, and bus) in the areas attributed with mixed land use. On the other hand, a higher land use mix may indicate a more pedestrian friendly infrastructure adhering to highly commercialized and smaller blocks, which would encourage pedestrian activities. This is reflected in the remaining 96.02% where the association is positive (Fig. 3(b)). Similarly, mixed land use increases the probability of choosing bicycling for 96.88% and 92.47% of the riders, in the context of access and egress trips, respectively. For the remaining 3.12% and 7.53%, the probability of choosing bicycling decreases (Fig. 3(c) and (d)). The explanation may be similar to that for walking.

5.1.3. Street design

Street design characteristics including road density and number of intersections are not found to play an important role in both transfer distance and choice probability for active modes, as expected (El-Geneidy et al., 2014; Zhao and Li, 2017; Chan and Farber, 2020). The results in Tables 3 and 4 demonstrate that road density is not significantly associated with either active mode distances or choice probability. Using the mean and standard deviation we find that for 63.71% of the riders, the association between road density and bicycling distance is positive. A higher road density in the area may indicate a better accessibility and connectivity, which may stimulate bicycling. For the remaining 36.29%, the association is in the opposite direction (Fig. 2(d)). A possible explanation is that higher road density may also result in heavier traffic and more frequent bus service.

In terms of number of intersections, a significant positive relationship is found with the probability of bicycling. The elasticity indicates that a 1% increase in the number of intersections will increase the probability of bicycling by 0.38% and 0.41%, for access and egress trips respectively. The relationship is not significant for the probability of walking. This result is consistent with Ding et al. (2017). It seems that well-connected neighborhoods better accommodate bicycling and decrease the probability of driving or taking a

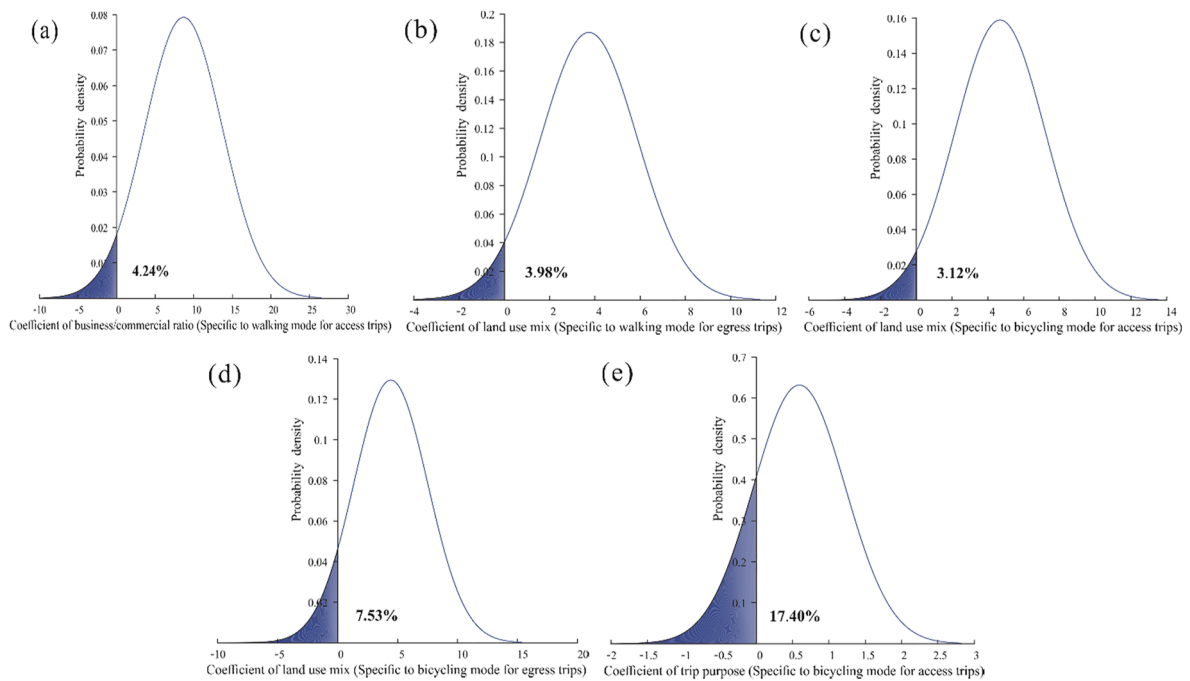


Fig. 3. Estimated parameter distributions of variables in transfer modes choice modelling.

bus.

5.1.4. Accessibility to bus

Interestingly, it was shown that the number of bus lines, a proxy for bus access, plays an important role in walking and bicycling transfer patterns. The results demonstrate that the number of bus lines is significantly associated with walking distance, but not with the probability of choosing walking, while the relationships with bicycling are reverse. It indicates that accessibility to bus influences the walking-metro integration mainly from the transfer distance perspective, whereas bicycling-metro integration is generally affected from the mode choice probability perspective.

The significant negative coefficient of the number of bus lines in Table 3 suggests that the presence of more bus lines decreases the walking distance of access/egress trips. The elasticity shows that a 1% increase in the number of bus lines will decrease walking distance by 0.96%. One of the main reasons might be that metro station areas with many bus lines provide more travel options, thus resulting in more pedestrians choosing a nearby bus stop than a distant metro station. The modelling and elasticity analysis results shown in Table 4 suggest that metro station areas with a larger number of bus lines decrease the probability of bicycling for access and egress trips, consistent with Zhao and Li (2017) and Liu et al. (2020a). Such results are plausible, considering that bus substitutes the bicycle as a transfer mode and a portion of integrated bicycle-metro use shifts to integrated bus-metro use. The number of bus lines is not significantly associated with the probability of choosing walking, both for access and egress trips.

5.2. Perceived walkability and bikeability

5.2.1. Perceived walkability

The presence of good walking facilities and an enjoyable walking environment are negatively associated with walking distance. It is somewhat surprising and counterintuitive. However, it should be kept in mind that the coefficients just reflect revealed relationships between people's self-reported assessments on walkability and transfer distance. This indicates that pedestrians with long access/egress trips are more likely to report a negative assessment of current walking facilities and walking environment. A plausible reason is that walking facilities (e.g., pedestrian crossing and sidewalk ground tile) and the environment surrounding metro stations are normally superior to walking facilities in other areas². People who walk a longer way are more likely to encounter less favorable walking facilities and environments such as discontinuous sidewalks and intersections without pedestrian signals. On the other hand, long access/egress distances more likely occur around suburban stations, because metro station density in outskirts is sparse and the catchment areas of these stations are larger compared to city center stations (Jiang et al., 2012; Wang and Cao, 2017).

² Actually, we divided the sample into two subgroups based on the access/egress distance (e.g., ≤ 800 m vs. > 800 m) and conducted *t*-test for these walking facility and environment variables. The results show that the differences are significant, indicating that the walking facilities and environment is not evenly distributed in the city.

As expected, after controlling for the built environment, trip attributes and socioeconomic characteristics, nearly all perceived walkability variables have significant positive effects on the probability of choosing walking, regardless of whether it is an access or egress trip. For example, the elasticities indicate that the probability of walking increases by 1.65% and 1.75% for access and egress trips respectively, when the evaluation of the presence of good walking facilities increases by 1%. Similarly, it increases by 0.94% and 1.28% when the evaluation of enjoyable walking environment increases by 1%.

5.2.2. Perceived bikeability

In terms of perceived bikeability, it appears that only enjoyable bicycling environment is significantly related to bicycling distance. The significant and negative coefficient indicates that the evaluations of the riders with long access/egress bicycling trips tend to be lower than the evaluations of people with a shorter bicycling distance. The potential reasons may be similar to those for walking. The relationships between the presence of good bicycling facilities, safe & enough parking space and bicycling distance are insignificant.

Conversely, the results in Table 4 demonstrate that all perceived bikeability factors are significantly associated with the probability of choosing bicycling, which is consistent with the literature (e.g., de Souza et al., 2017; Zhao and Li, 2017; Wu et al., 2018). For example, the elasticities show that a 0.88% and 1.36% (for access and egress trips, respectively) increase in choosing bicycling follows a 1% increase in the evaluation of safe & enough parking space. Comparing the elasticities of these three independent variables, it is clear that the influence of the presence of good bicycling facilities is strongest, followed by safe & enough parking space, and enjoyable bicycling environment. It indicates that bicyclists are most concerned about bicycling infrastructure. Compared to enhancing the other two perceived bikeability factors, providing good bicycling facilities has most potential to promote the probability of choosing bicycling as the transfer mode in conjunction with metro.

5.3. Trip attributes

The results in Table 3 show that, in the context of access trips, walking and bicycling distance could be increased by 30.63% and 34.20%, respectively. Although the relationships are modest as indicated by the signs of the coefficients, the results show a negative association between the dummy variable ‘urban’ and walking/bicycling distance, and a positive relationship with the probability of active mode choice. This is consistent with the findings of El-Geneidy et al. (2014), Wang and Cao (2017), and Guo and He (2020), that metro station density in urban areas is normally higher than in suburban areas. Consequently, shorter walking distances are needed in urban areas for metro access and egress. As shown in Table 3, walking and bicycling distance decline by 20.16% and 3.29% when the metro station is located in the urban area. The parameter of the access indicator is normally distributed with mean 0.113 and standard deviation 0.130 in the walking distance model, and mean 0.550 and standard deviation 0.951 in the bicycling distance model. It indicates that for a minority of individuals, access walking and bicycling distance is shorter than egress walking and bicycling distance (19.32% and 28.16%, see Fig. 2(c) and (e)).

Peak hours play a trivial role in transfer distance and choice of walking and bicycling, as indicated by the corresponding signs of the coefficients. Travel purpose is not significantly associated with transfer distance and the probability of choosing walking. However, it is significantly associated with the probability of choosing bicycling. Compared to the people travelling for leisure and shopping (flexible travel demand), commuters (rigid travel demand) are prone to choosing bicycles as the transfer mode for metro. This may be due to the pressing need to get to work and school, so that travelers tend to choose an efficient and reliable mode. In addition, it is found that the estimated parameter for trip purpose passes the significance test for a normal distribution (0.593, 0.632) in the access trip model. It indicates that most travelers (82.60%) prefer to choose bicycling for utilitarian purposes (i.e., go to work or school), whereas a small proportion of travelers (17.40%) have lower probability of choosing bicycling for utilitarian purposes relative to other travel purposes

Table 5

Summary of the associations between perceived walkability/bicycle-ability, built environment and transfer patterns.

Variables	Access/egress distance of active modes	Probability of choosing active modes
Built environment		
Population density	/	+ [W]
Residential ratio	/	- [W]
Business/commercial ratio	/	+ [W]
Industrial/manufacturing ratio	- [W]	/
Land use mix	+ [W]	+ [W&B]
Number of intersections	/	+ [B]
Road density	/	/
Number of bus lines	- [W]	- [B]
Perceived walkability/bikeability		
Presence of good walking facilities	- [W]	+ [W]
Ease of walking	/	+ [W]
Enjoyable walking environment	- [W]	+ [W]
Presence of good riding facilities	/	+ [B]
Enjoyable riding environment	- [B]	+ [B]
Safe & enough parking space	/	+ [B]

Notes: ‘+’ significant positive relationship, ‘-’ significant negative relationship, ‘/’ insignificant relationship; variables in bold denotes significance with both distance and choice probability; [W] denotes significant for walking, [B] for bicycling, and [W&B] for both.

(i.e., recreation) (see Fig. 3(e)).

6. General discussion

Compared to bicycling distance, walking distance is more sensitive to travelers' perceived walkability and the built environment since more factors are significant in the walking distance model than in the bicycling distance model. For example, presence of good walking facilities is significantly associated with walking distance, whereas presence of good riding facilities has an insignificant effect on bicycling distance. It indicates that travelers are more sensitive to walkability. The results show that the built environment plays a trivial role in bicycling distance as all effects of features of the built environment (i.e., industrial/manufacturing ratio, land use mix, and number of bus lines) are insignificant. The negative relationship between perceived walkability/bikeability and transfer distance of active travel modes indicates that one should be very cautious interpreting estimated parameters of revealed preference model in behavioral terms. Often the estimated effects are mainly an artifact of the specific spatial distributions and corresponding correlations of the dependent and independent variables and have no behavioral meaning. Moreover, the behavior of a substantial part of our sample is constrained. In this case, the estimated coefficients do not manifest that increasing perceived walkability/bikeability decreases walking and bicycling distances in the context of metro access/egress trips, but reflect the fact that travelers with longer walking and bicycling distances tend to be less satisfied with some dimensions of current walkability. It possibly indicates that walkability/bikeability in distant metro catchment areas is worse than in areas near to metro stations.

Table 3 shows that transfer stage (access or egress) is significantly associated with both walking and bicycling distances and indicates that walking and bicycling access distances tend to be longer than walking and bicycling egress distances. However, the effect directions of the specific perceived walkability/bikeability and built environment variables on access and egress mode choice are the same and the differences between the magnitudes of the parameter estimates are small (Table 4), indicating that the disparities of the effects of perceived walkability/bikeability and the built environment on mode choice between access and egress trips are small.

We summarize the relationships between the built environment, perceived walkability/bikeability, and transfer patterns in Table 5 based on the estimation results shown in Tables 3 and 4. Estimating the dependence between transfer distance and mode choice behavior for active modes (walking and bicycling) illustrates interesting results. It shows that more factors of perceived walkability/bikeability and the built environment are associated with mode choice than with distance. This indicates that although travelers' perceptions of walkability/bikeability and the built environment play a role in the use of active travel modes in the context of metro access/egress trips from both distance and choice probability perspectives, promoting the integration of active mode and metro is more workable and effective in the way of increasing the probability of choosing walking and bicycling than in the way of expanding walking and bicycling distance.

7. Conclusions

The integration of active modes and public transit has attracted attention from researchers and policymakers alike from across the world as an efficient and sustainable urban transportation system, which contributes to reducing car usage and the development of green and sustainable cities. Although there is a growing interest in transfer patterns and investigating the impact of the built environment on these patterns, few studies have considered both distance and mode choice behavior in a single study. Furthermore, few studies have examined bicycling distance as well as walking distance and compared the possible differential associations of the built environment and these distances in the context of metro transfer trips. We therefore attempted to scrutinize the associations of perceived walkability/bikeability, built environment and transfer patterns (walking/bicycling distance and mode choice probability), using survey data from Nanjing. Findings of this study contribute to deepen our understanding on the associations of perceived walkability/bikeability, built environment and transfer distances of active modes versus their choice probabilities, and maximize the integration of active modes and the metro system.

A number of key findings of this study and corresponding recommendations are worth to be summarized and highlighted. First, travelers' decisions of choosing an active metro access/egress mode than of walking or bicycling a longer way to/from metro stations are more likely to be associated with perceived walkability/bikeability and the built environment. It indicates that stimulating more people in the catchment areas to choose active modes as metro transfer modes is probably a more feasible and vigorous way than working on expanding the metro catchment areas for promoting the integration of active modes and metro system by changing the built environment and perceived walkability/bikeability. Second, in Nanjing, providing good walkability/bikeability as well as altering the built environment is an important approach for improving the integration of active modes and metro systems. Urban planners and policymakers should pay enough attention to the walkability and bikeability details such as pedestrian crossings, smooth sidewalk ground tiles, continuous sidewalks, separate bicycle lanes, tree shade, and eye-level greenness, avoiding barriers for the use of active modes for access/egress trips (de Souza et al., 2017). Third, similar to the findings of Sun et al. (2017), the present study suggests that features of the built environment may not show a significant relationship with one aspect of transfer patterns (i.e., distance) but is significantly related to another aspect such as mode choice behavior. Thus, changing the built environment may only affect one aspect or may even have adverse effects of different aspects of transfer patterns. Therefore, analyses of different aspects of transfer patterns of metro access/egress trips provide a more comprehensive and appropriate approach to improving the integration of metro and active transfer modes in practice (Yang et al., 2019; van Soest et al., 2020). Fourth, transfer stage seems to play an important role in walking and bicycling distances, while the differences in transfer mode choice behavior between access and egress trips seem small.

In terms of the effect of specific features of the built environment, the proper distribution of urban land use can not only enlarge the metro catchment areas but also increase people's willingness to choose walking/bicycling for transfer trips. Mixed land use

development should be encouraged, which echoes the findings of Zhao and Li (2017) and Ding et al. (2018). Residential ratio and industrial/manufacturing ratio are negatively associated with the likelihood of choosing walking and walking distance. Considering the reality that the residential communities and factories in Chinese cities are normally centralized, large and enclosed and in consequence produce distant detours and reduce accessibility, policy implications such as small blocks and open communities should be addressed in urban planning. The negative relationship between walking distance and number of bus lines indicates that there is an obvious substitution effect between metro and bus. The presence of more bus lines decreases the likelihood of choosing bicycling towards to/from metro stations suggesting that bus may substitute the bicycle as a transfer mode. A decrease in the number of bus lines (or bus stops) in the metro catchment areas may promote the integration of active transfer modes and metro, but compared to the dense bus stops, sparse metro stations cannot meet people's intricate spatial mobility demand (Zhao and Li, 2017; Wang et al., 2020). Moreover, the inconvenience and higher costs³ of public transportation may result in travelers' avoiding public transportation and even stimulate their desire to purchase cars. Therefore, it raises a dilemma which needs further exploration and discussion. Therefore, policymakers should be very careful in arranging bus stops/lines in metro catchment areas. They should take not only the competitiveness of buses but also the charm of public transit into account.

This study has some limitations, mostly rooted in data availability constraints, which at the same time provide possible avenues for future research. First, the study is cross-sectional in nature and therefore the estimated coefficients should not be interpreted as causal effects. Establishing the causal mechanisms underlying the observed association between the independent variables and metro transfer patterns ideally requires additional longitudinal data and research methods that allow causal inference. Second, even though we controlled the socioeconomic characteristics and trip attributes (e.g., travel purpose, access/egress distance) in the model, variables such as weather information and level-of-service attributes (i.e., access/egress time, monetary access/egress cost for passive transport modes) are not addressed in the present study due to data limitations. The omission of these variables may lead to biased estimates of the calibrated travel choice models (Ben-Akiva and Lerman, 1985). This issue could be alleviated with more comprehensive behavioral data, which could produce a better understanding of the integration of active modes and the metro system. Third, the present study has been able to demonstrate how much residents like walk or bike to and from metro station, but not which other options are available to them. Given that the bus also plays a key role as a main transportation mode, knowing more about bus mode choice as a main as well as access/egress mode can enrich the analysis, and thus help better understanding travel behavior in a certain city. Additionally, due to the small number of observations, we grouped the replaced transport modes by active access/egress modes (i.e., taxi, ride-hailing, bus) into one single group of passive transport modes, ignoring the intra-group heterogeneity. This aspect could be further improved by gathering more data in the future. Finally, the associations between the built environment and metro transfer patterns extracted from Nanjing, China may not be transferable to other cities due to geographical and cultural differences. It will be meaningful to extend the study to other cities to examine the transferability of empirical knowledge.

CRedit authorship contribution statement

Zuoxian Gan: Data collection, Methodology, Investigation, Writing – original draft. **Min Yang:** Conceptualization, Project administration, Funding acquisition. **Qingcheng Zeng:** Formal analysis. **Harry J.P. Timmermans:** Methodology, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Arellana, J., Saltarín, M., Larrañaga, A.M., Alvarez, V., Henao, C.A., 2020. Urban walkability considering pedestrians' perceptions of the built environment: a 10-year review and a case study in a medium-sized city in Latin America. *Transp. Rev.* 40 (2), 183–203.
- Aziz, H.M.A., Nagle, N.N., Morton, A.M., Hilliard, M.R., White, D.A., Stewart, R.N., 2018. Exploring the impact of walk–bike infrastructure, safety perception, and built-environment on active transportation mode choice: a random parameter model using New York City commuter data. *Transportation* 45 (5), 1207–1229.
- Ben-Akiva, M., Lerman, S.R., 1985. *Discrete Choice Analysis: Theory and Application to Travel Demand*. MIT Press, Cambridge MA, USA.

³ Metro fare is normally higher than bus fare in Chinese cities. Take Nanjing as an example, the starting price for metro system is 2 *yuan* with 4 km, and the price increases with different scales with increasing trip distances and up to 15 *yuan*; whereas, fixed fare is adopted in bus system, and the price is 1 *yuan* for non-air-conditioned bus and 2 *yuan* for air-conditioned bus, respectively.

- Böcker, L., Anderson, E., Uteng, T.P., Throndsen, T., 2020. Bike sharing use in conjunction to public transport: exploring spatiotemporal, age and gender dimensions in Oslo, Norway. *Transp. Res. Part A: Policy Practice* 138, 389–401.
- Brons, M., Rietveld, P., 2009. Improving the quality of the door-to-door rail journey: a customer-oriented approach. *Built Environ.* 35 (1), 122–135.
- Cervero, R., Kockelman, K., 1997. Travel demand and the 3Ds: density, diversity, and design. *Transp. Res. Part D: Transp. Environ.* 2 (3), 199–219.
- Chan, K., Farber, S., 2020. Factors underlying the connections between active transportation and public transit at commuter rail in the Greater Toronto and Hamilton Area. *Transportation* 47 (5), 2157–2178.
- de Souza, F., La Paix Puello, L., Brussel, M., Orrico, R., van Maarseveen, M., 2017. Modelling the potential for cycling in access trips to bus, train and metro in Rio de Janeiro. *Transp. Res. Part D: Transp. Environ.* 56, 55–67.
- Ding, C., Wang, D., Liu, C., Zhang, Y.i., Yang, J., 2017. Exploring the influence of built environment on travel mode choice considering the mediating effects of car ownership and travel distance. *Transp. Res. Part A: Policy Practice* 100, 65–80.
- Ding, C., Wang, Y., Tang, T., Mishra, S., Liu, C., 2018. Joint analysis of the spatial impacts of built environment on car ownership and travel mode choice. *Transp. Res. Part D: Transp. Environ.* 60, 28–40.
- El-Geneidy, A., Grimsrud, M., Wasfi, R., Tétreault, P., Surprenant-Legault, J., 2014. New evidence on walking distances to transit stops: identifying redundancies and gaps using variable service areas. *Transportation* 41 (1), 193–210.
- Ewing, R., Cervero, R., 2010. Travel and the built environment: a meta-analysis. *J. Am. Plan. Assoc.* 76 (3), 265–294.
- Fernández-Heredia, Á., Jara-Díaz, S., Monzón, A., 2016. Modelling bicycle use intention: the role of perceptions. *Transportation* 43 (1), 1–23.
- Gan, Z., Feng, T., Yang, M., 2019. Exploring the effects of car ownership and commuting on subjective well-being: a nationwide questionnaire study. *Sustainability* 11 (1), 84. <https://doi.org/10.3390/su11010084>.
- Gan, Z., Yang, M., Feng, T., Timmermans, H.J.P., 2020. Examining the relationship between built environment and metro ridership at station-to-station level. *Transp. Res. Part D: Transp. Environ.* 82, 102332. <https://doi.org/10.1016/j.trd.2020.102332>.
- Guo, Y., He, S.Y., 2020. Built environment effects on the integration of dockless bike-sharing and the metro. *Transp. Res. Part D: Transp. Environ.* 83, 102335. <https://doi.org/10.1016/j.trd.2020.102335>.
- He, J., Zhang, R., Huang, X., Xi, G., 2018. Walking access distance of metro passengers and relationship with demographic characteristics: a case study of Nanjing Metro. *Chin. Geo. Sci.* 28 (4), 612–623.
- Hochmair, H.H., 2015. Assessment of bicycle service areas around transit stations. *Int. J. Sustain. Transp.* 9 (1), 15–29.
- Jamal, S., Mohiuddin, H., Paez, A., 2020. How do the perceptions of neighborhood conditions impact active transportation? A study in Rajshahi, Bangladesh. *Transp. Res. Part D: Transp. Environ.* 87, 102525. <https://doi.org/10.1016/j.trd.2020.102525>.
- Ji, Y., Fan, Y., Ermagun, A., Cao, X., Wang, W., Das, K., 2017. Public bicycle as a feeder mode to rail transit in China: the role of gender, age, income, trip purpose, and bicycle theft experience. *Int. J. Sustain. Transp.* 11 (4), 308–317.
- Jiang, Y., Christopher Zegers, P., Mehndiratta, S., 2012. Walk the line: station context, corridor type and bus rapid transit walk access in Jinan. China. *J. Transp. Geogr.* 20 (1), 1–14.
- La Paix Puello, L., Geurs, K.T., 2015. Modelling observed and unobserved factors in cycling to railway stations: application to transit-oriented-developments in the Netherlands. *Eur. J. Transp. Infrastruct. Res.* 15, 27–50.
- Li, S., Zhao, P., Zhang, H.c., Quan, J., 2019. Walking behavior in the old downtown Beijing: The impact of perceptions and attitudes and social variations. *Transp. Policy* 73, 1–11.
- Li, X., Tang, J., Hu, X., Wang, W., 2020. Assessing intercity multimodal choice behavior in a Touristy City: a factor analysis. *J. Transp. Geogr.* 86, 102776. <https://doi.org/10.1016/j.jtrangeo.2020.102776>.
- Lin, D., Zhang, Y., Zhu, R., Meng, L., 2019. The analysis of catchment areas of metro stations using trajectory data generated by dockless shared bikes. *Sustain. Cities Society* 49, 101598. <https://doi.org/10.1016/j.scs.2019.101598>.
- Lin, J.-J., Zhao, P., Takada, K., Li, S., Yai, T., Chen, C.-H., 2018. Built environment and public bike usage for metro access: a comparison of neighborhoods in Beijing, Taipei, and Tokyo. *Transp. Res. Part D: Transp. Environ.* 63, 209–221.
- Liu, Y., Ji, Y., Feng, T., Timmermans, H., 2020a. Understanding the determinants of young commuters' metro-bikeshare usage frequency using big data. *Travel Behav. Society* 21, 121–130.
- Liu, Y., Yang, D., Timmermans, H.J.P., de Vries, B., 2020b. Analysis of the impact of street-scale built environment design near metro stations on pedestrian and cyclist road segment choice: a stated choice experiment. *J. Transp. Geogr.* 82, 102570. <https://doi.org/10.1016/j.jtrangeo.2019.102570>.
- Nanjing Statistical Bureau, 2019. 2019' Nanjing Statistical Yearbook. China Statistic Press, Nanjing <http://tjj.nanjing.gov.cn/>.
- Nielsen, T.A.S., Skov-Petersen, H., 2018. Bikeability-Urban structures supporting cycling. Effects of local, urban and regional scale urban form factors on cycling from home and workplace locations in Denmark. *J. Transp. Geogr.* 69, 36–44.
- Ma, L., Ettema, D., Ye, R., 2021. Determinants of bicycling for transportation in disadvantaged neighbourhoods: evidence from Xi'an, China. *Transp. Res. Part A: Policy Practice* 145, 103–117.
- Ma, X., Ji, Y., Jin, Y., Wang, J., He, M., 2018. Modeling the factors influencing the activity spaces of bikeshare around metro stations: a spatial regression model. *Sustainability* 10 (11), 3949. <https://doi.org/10.3390/su10113949>.
- Park, S., Kang, J., Choi, K., 2014. Finding determinants of transit users' walking and biking access trips to the station: a pilot case study. *KSCE J. Civ. Eng.* 18 (2), 651–658.
- Rietveld, P., 2000. Non-motorised modes in transport systems: a multimodal chain perspective for The Netherlands. *Transp. Res. Part D: Transp. Environ.* 5 (1), 31–36.
- Shelat, S., Huisman, R., van Oort, N., 2018. Analysing the trip and user characteristics of the combined bicycle and transit mode. *Res. Transp. Econ.* 69, 68–76.
- Sun, B., Ermagun, A., Dan, B.o., 2017. Built environmental impacts on commuting mode choice and distance: evidence from Shanghai. *Transp. Res. Part D: Transp. Environ.* 52, 441–453.
- Sun, G., Zacharias, J., Ma, B., Oreskovic, N.M., 2016. How do metro stations integrate with walking environments? Results from walking access within three types of built environment in Beijing. *Cities* 56, 91–98.
- Tao, T., Wang, J., Cao, X., 2020. Exploring the non-linear associations between spatial attributes and walking distance to transit. *J. Transp. Geogr.* 82, 102560. <https://doi.org/10.1016/j.jtrangeo.2019.102560>.
- van Soest, D., Tight, M.R., Rogers, C.D.F., 2020. Exploring the distances people walk to access public transport. *Transp. Rev.* 40 (2), 160–182.
- Wang, J., Cao, X., 2017. Exploring built environment correlates of walking distance of transit egress in the Twin Cities. *J. Transp. Geogr.* 64, 132–138.
- Wang, R., Lu, Y.i., Wu, X., Liu, Y.e., Yao, Y., 2020. Relationship between eye-level greenness and cycling frequency around metro stations in Shenzhen, China: a big data approach. *Sustain. Cities Society* 59, 102201. <https://doi.org/10.1016/j.scs.2020.102201>.
- Washington, S., Karlaftis, M., Mannering, F., 2010. *Statistical and Econometric Methods for Transportation Data Analysis*. CRC Press, New York.
- Wu, J., Yang, M., Sun, S., Zhao, J., 2018. Modeling Travel Mode Choices in Connection to Metro Stations by Mixed Logit Models: a Case Study in Nanjing, China. *Promet – Traf. & Transp.* 30 (5), 549–561.
- Yang, H., Dijst, M., Feng, J., Ettema, D., 2019. Mode choice in access and egress stages of high-speed railway travelers in China. *J. Transp. Land Use* 12 (1), 701–721.
- Zacharias, J., Zhao, Q.i., 2018. Local environmental factors in walking distance at metro stations. *Publ. Transp.* 10 (1), 91–106.
- Zhao, P., Li, S., 2017. Bicycle-metro integration in a growing city: the determinants of cycling as a transfer mode in metro station areas in Beijing. *Transp. Res. Part A: Policy Practice* 99, 46–60.